SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I Dhivya Dharshini S *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dhivya

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q_5
Understanding of Scenario	20%	1	1	1	1	1
Identification of Key Issues	20%	1	1	1	1	1
Application of Concepts	25%	2	2	2	2	2
Decision Making	20%	1	1	1	1	1
Clarity of Response	15%	1	0	1	0	0

27/30

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

ANSWER

Q1. Smart Campus: 192.168.100.0/24 for Computing (/25), Electronics (/26), Mechanical (/27) (6 marks)

1. Scenario understanding

- **Given:** the block 192.168.100.0/24 (256 addresses) and fixed masks: School of Computing /25, School of Electronics /26, School of Mechanical Engineering /27.
- **Required:** the network address, broadcast address, valid host range and usable hosts for each school, plus the unused range left in the block.

2. Key issues

- The three subnets have different sizes (128, 64 and 32 addresses), so they must be placed without overlapping.
- Each subnet must start on a multiple of its own block size.
- The first and last address of every subnet are reserved for the network and broadcast, so usable hosts = 2^h − 2.

3. Concepts applied

School	Prefix	Dotted mask	Host bits	Total addresses	Block size
Computing	/25	255.255.255.128	7	$2^7 = 128$	128
Electronics	/26	255.255.255.192	6	$2^6 = 64$	64
Mechanical	/27	255.255.255.224	5	$2^5 = 32$	32

- Broadcast = network + block size − 1. First host = network + 1. Last host = broadcast − 1.

4. Working (largest subnet first)

- **Computing:** network 192.168.100.0. Broadcast = 0 + 128 − 1 = 127. Hosts = .1 to .126. Usable = $2^7 − 2 =$ **126**.
- **Electronics:** starts at 127 + 1 = 128. Broadcast = 128 + 64 − 1 = 191. Hosts = .129 to .190. Usable = $2^6 − 2 =$ **62**.
- **Mechanical:** starts at 191 + 1 = 192. Broadcast = 192 + 32 − 1 = 223. Hosts = .193 to .222. Usable = $2^5 − 2 =$ **30**.
- **Unused:** the three subnets use 128 + 64 + 32 = 224 addresses (.0 to .223). The remainder is .224 to .255, which is 256 − 224 = 32 addresses.

5. Final answer

School	Subnet	Network	Valid host range	Broadcast	Usable hosts
Computing	192.168.100.0/25	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	**126**
Electronics	192.168.100.128/26	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.191	**62**
Mechanical	192.168.100.192/27	192.168.100.192	192.168.100.193 – 192.168.100.222	192.168.100.223	**30**

Unused range: 192.168.100.224 – 192.168.100.255 (32 addresses, i.e. 192.168.100.224/27).

6. Decision and justification

- Allocating the largest subnet first keeps every network address on its block boundary and leaves one clean free block at the end.
- The leftover /27 can be given to a new department later without renumbering the existing schools.
- Each school is its own broadcast domain, so a router or Layer 3 switch is needed between them. This also keeps their traffic separate for security.

Q2. Software company: 10.10.0.0/16 for Development (/24), QA (/25), HR (/26), Executive (/27) (6 marks)

1. Scenario understanding

- **Given:** the private block 10.10.0.0/16 (65,536 addresses) and masks Software Development /24, Quality Assurance /25, Human Resources /26, Executive Management /27.
- **Required:** network and broadcast addresses, valid host ranges, usable hosts for each department, and the unused range left in the /16.

2. Key issues

- The /24 subnet takes a whole third octet, so the smaller subnets start in the next block, 10.10.1.x.
- A /24 in this block changes the third octet, while /25, /26 and /27 subnet the fourth octet.
- No overlap is allowed, and the unused space must be reported as one continuous range.

3. Concepts applied

Dept	Prefix	Dotted mask	Host bits	Total addresses	Block size
Software Development	/24	255.255.255.0	8	$2^8 = 256$	256 (one third-octet value)
Quality Assurance	/25	255.255.255.128	7	$2^7 = 128$	128
Human Resources	/26	255.255.255.192	6	$2^6 = 64$	64
Executive Management	/27	255.255.255.224	5	$2^5 = 32$	32

4. Working (largest subnet first)

- **Software Development:** network 10.10.0.0. Broadcast = 10.10.0.255. Hosts = 10.10.0.1 to 10.10.0.254. Usable = $2^8 - 2$ = **254**.
- **Quality Assurance:** starts at 10.10.1.0. Broadcast = 0 + 128 − 1 = 127, so 10.10.1.127. Hosts = .1 to .126. Usable = $2^7 - 2$ = **126**.
- **Human Resources:** starts at 10.10.1.128. Broadcast = 128 + 64 − 1 = 191, so 10.10.1.191. Hosts = .129 to .190. Usable = $2^6 - 2$ = **62**.
- **Executive Management:** starts at 10.10.1.192. Broadcast = 192 + 32 − 1 = 223, so 10.10.1.223. Hosts = .193 to .222. Usable = $2^5 - 2$ = **30**.
- **Unused:** allocated = 256 + 128 + 64 + 32 = 480 addresses (10.10.0.0 to 10.10.1.223). Unused = 65,536 − 480 = **65,056 addresses**.

5. Final answer

Dept	Subnet	Network	Valid host range	Broadcast	Usable hosts
Software Development	10.10.0.0/24	10.10.0.0	10.10.0.1 – 10.10.0.254	10.10.0.255	**254**
Quality Assurance	10.10.1.0/25	10.10.1.0	10.10.1.1 – 10.10.1.126	10.10.1.127	**126**
Human Resources	10.10.1.128/26	10.10.1.128	10.10.1.129 – 10.10.1.190	10.10.1.191	**62**
Executive Management	10.10.1.192/27	10.10.1.192	10.10.1.193 – 10.10.1.222	10.10.1.223	**30**

Unused range: 10.10.1.224 – 10.10.255.255 (65,056 addresses). The first free block is 10.10.1.224/27.

6. Decision and justification

- The departments are placed from largest to smallest, so they sit back to back in the first 480 addresses with no gaps.
- Development gets the biggest subnet (254 hosts) because it has the most employees. Executive Management needs only a /27 (30 hosts).
- About 99.3% of the /16 is still free (65,056 of 65,536), so the company can add departments or buildings without redesigning the plan.

Q3. Smart hospital: IPv6 network 2001:0db8:abcd:0000:0000:0000:0000:0000/64 (6 marks)

1. Scenario understanding

- **Given:** the IPv6 network 2001:0db8:abcd:0000:0000:0000:0000:0000 with prefix length /64, to serve thousands of medical devices and IoT sensors.
- **Required:** the compressed form, the expanded form, the /64 network prefix, and the number of host addresses in the subnet.

2. Key issues

- An IPv6 address is 128 bits written as 8 groups of 16 bits, so the rules for dropping zeros and using :: must be applied correctly.
- :: may replace only one run of all-zero groups, and it must be the longest run.
- The /64 split gives 64 network bits and 64 host (interface) bits.

3. Concepts applied

Group	1	2	3	4	5	6	7	8
Full form	2001	0db8	abcd	0000	0000	0000	0000	0000
Leading zeros dropped	2001	db8	abcd	0	0	0	0	0
Part of /64	Prefix	Prefix	Prefix	Prefix	Interface ID	Interface ID	Interface ID	Interface ID

4. Working

Compression:

1. Drop leading zeros: 2001:db8:abcd:0:0:0:0:0.
2. The longest run of zero groups is groups 4 to 8 (five groups).
3. Replace that run with ::. Compressed form: **2001:db8:abcd::/64**.

Expansion:

1. 2001:db8:abcd:: shows 3 groups, so :: stands for $8 - 3 = 5$ groups of 0000.
2. Pad every group to 4 digits: **2001:0db8:abcd:0000:0000:0000:0000:0000/64**. This matches the given address.

Network prefix (/64): 64 bits = the first 4 groups: 2001:0db8:abcd:0000. The prefix is **2001:db8:abcd::/64** (also written 2001:db8:abcd:0::/64).

Host addresses: host bits = $128 - 64 = 64$, so the total is 2^{64} = **18,446,744,073,709,551,616** addresses. This is the answer to both of the last two bullets in the question, which ask for the same number.

5. Final answer

Item	Result
Compressed notation	2001:db8:abcd::/64
Expanded notation	2001:0db8:abcd:0000:0000:0000:0000:0000/64
Network prefix	2001:db8:abcd::/64
Host bits	64
Host addresses	2^{64} = 18,446,744,073,709,551,616

6. Decision and justification

- A /64 is the standard IPv6 subnet size, so the hospital can use one /64 per ward or device group and still have billions of billions of addresses each.
- Thousands of medical devices and IoT sensors use only a tiny part of the 2^{64} addresses, so address shortage (the IPv4 problem) no longer exists.

- IPv6 has no broadcast address, so essentially all addresses are usable. The given address has an all-zeros interface ID, so it names the network itself (the subnet-router anycast address) and should not be given to a device.

Q4. ISP core router: forwarding a packet to 192.168.2.45 (6 marks)

1. Scenario understanding

- **Given:** a core router with the routes 192.168.1.0/24, 192.168.2.0/24 and 192.168.3.0/24, and a packet with destination 192.168.2.45.
- **Required:** the destination subnet, the matching routing table entry, the next hop or interface, and the default route for unmatched destinations.

2. Key issues

- The router compares the destination IP with each route using the route's own mask. This is called **longest prefix match**.
- All three routes are /24, so the first 24 bits (the first three octets) are compared.
- The table shows networks only. The next hop and interface names are not listed, so they are described by the route they belong to.

3. Concepts applied

- Destination network = destination IP AND mask (255.255.255.0).
- The packet is sent out of the interface or next hop of the route that matches. If nothing matches, the **default route 0.0.0.0/0** is used (the gateway of last resort).

4. Working

The third octet decides the match, since the first two octets (192.168) are the same in all routes.

Route	Third octet (binary)	192.168.2.45 AND 255.255.255.0	Match?
192.168.1.0/24	00000001	192.168.2.0	No, $2 \neq 1$
192.168.2.0/24	00000010	192.168.2.0	**Yes**
192.168.3.0/24	00000011	192.168.2.0	No, $2 \neq 3$

- The destination subnet is 192.168.2.0/24: network 192.168.2.0, valid hosts 192.168.2.1 – 192.168.2.254, broadcast 192.168.2.255. 192.168.2.45 is a valid host in it.

5. Final answer

Item	Result
Destination subnet	192.168.2.0/24
Matching routing table entry	192.168.2.0/24 (the only route that matches)
Next hop / interface	The interface or next-hop router that the table lists for 192.168.2.0/24, i.e. the link to customer network 2
Default route	0.0.0.0/0, pointing to the ISP's upstream router (gateway of last resort)

6. Decision and justification

- Exactly one route matches, so there is no conflict and no need to compare prefix lengths.
- The packet is forwarded only towards customer network 2, so customers 1 and 3 do not receive it. This keeps customer traffic separate.
- A packet for a destination such as 192.168.9.5 matches none of the three routes, so the router would use 0.0.0.0/0. Without a default route, it would drop that packet and send an ICMP destination-unreachable message.

Q5. Two LANs joined by a Layer-3 router: 192.168.10.0/26 and 192.168.10.64/26 (6 marks)

1. Scenario understanding

- **Given:** Administration LAN 192.168.10.0/26 with router interface 192.168.10.1, and Finance LAN 192.168.10.64/26 with router interface 192.168.10.65.
- **Required:** network and broadcast addresses, valid host ranges, three suitable computer addresses per LAN, and the default gateway for each LAN.

2. Key issues

- /26 means 255.255.255.192, so each LAN has a block of 64 addresses and $2^6 - 2 = 62$ usable hosts.
- Computer addresses must be inside the LAN's host range, must not equal the network, broadcast or gateway address, and must not repeat.
- Computers in different LANs can talk only through the router, so each computer needs the gateway of its own LAN.

3. Concepts applied

- Broadcast = network + 64 − 1. Host range = network + 1 to broadcast − 1.
- The default gateway of a host is the router interface in the host's own subnet.

4. Working

- **Administration LAN:** network 192.168.10.0. Broadcast = 0 + 63 = 63, so 192.168.10.63. Hosts = .1 to .62.
- **Finance LAN:** network 192.168.10.64. Broadcast = 64 + 63 = 127, so 192.168.10.127. Hosts = .65 to .126.
- The gateways .1 and .65 are the first host addresses of their LANs, so both lie inside the valid ranges.
- Computers take the next free addresses after the gateway.

5. Final answer

Item	Administration LAN	Finance LAN
Subnet / mask	192.168.10.0/26 (255.255.255.192)	192.168.10.64/26 (255.255.255.192)
Network address	192.168.10.0	192.168.10.64
Broadcast address	192.168.10.63	192.168.10.127
Valid host range	192.168.10.1 – 192.168.10.62	192.168.10.65 – 192.168.10.126
Usable hosts	62	62
Computer 1	192.168.10.2	192.168.10.66
Computer 2	192.168.10.3	192.168.10.67
Computer 3	192.168.10.4	192.168.10.68
Default gateway	**192.168.10.1**	**192.168.10.65**

Each computer uses the mask 255.255.255.192.

6. Decision and justification

- The addresses .2 to .4 and .66 to .68 are valid hosts that avoid the gateway, the network address and the broadcast address.
- The Administration computers use 192.168.10.1 and the Finance computers use 192.168.10.65, because a host sends traffic for other subnets to the router interface on its own LAN.
- Example: when 192.168.10.2 sends to 192.168.10.66, the destination is in another subnet, so the packet goes to 192.168.10.1. The router then forwards it out of its Finance interface (192.168.10.65).
- The two /26 blocks do not overlap (.0–.63 and .64–.127), and .128–.255 is still free for another LAN.

RUBRICS FOR CALCULATION

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